

FIRST-ORDER MATCHING AND PRECISION COSTS UNDER NONCOLLAPSING CONTROL MAPS

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ABSTRACT. We study the minimum size of an open-loop input catalogue that keeps every assigned actual initial state within a tolerance of a triangular constraint at all times of a finite horizon. Two planar C^2 control maps make the constraint controlled invariant, contract toward a common origin, and have critical sets of area zero in the constraint. Their first derivatives have angular branch proportions $a_0 + a_1 = 1$ and radial factors $r_i = a_i^{\beta_i}$, $\beta_i > 1$. Within this family, all positive-area compact initial sets share a cost rate on a positive precision interval adjacent to zero if and only if $\beta_0 = \beta_1$. For $\delta_n \rightarrow 0$ with $\delta_n = \exp(-\gamma n + o(n))$, matching with power β gives the full-constraint spectrum $\max_p H_b(p) \min\{1, \gamma/[\beta\chi(p)]\}$, where H_b is binary entropy and $\chi(p) = -p \log a_0 - (1-p) \log a_1$. Its common-cost and saturation boundaries are βh and $-(\beta/2) \log(a_0 a_1)$, with $h = -\sum_i a_i \log a_i$. One fixed compact set of arbitrarily nearly full area has rate $\min\{h, \gamma/\beta\}$ simultaneously for all finite exponents and their precision sequences. Under mismatch, the full-constraint rate is γD and one fixed nearly full-area compact set has the strictly smaller rate $\gamma h/\chi_R$ for $0 < \gamma < -\log \max_i r_i$; here $r_0^D + r_1^D = 1$ and $\chi_R = -\sum_i a_i \log r_i$. An actual-area estimate for every approximate input handles exit, reentry and truncated local feasible images. A regular compact window controls all finite transient inverse branches. Physical bounded-run trajectories and stopping counts give the matching catalogue bounds. A smooth feasible fold shows that global injectivity is unnecessary.

1. INTRODUCTION

Invariance entropy measures the number of control signals needed to maintain a state constraint; see Kawan [8]. When every input contracts toward a common point, a finite catalogue can suffice for arbitrarily long horizons at each fixed positive tolerance. The catalogue can nevertheless grow exponentially when the tolerance decreases exponentially with the horizon. We study this joint limit and ask whether its cost is shared by all compact initial sets of positive area.

The distinction between typical initial states and the full constraint is essential. A compact set of almost full area can omit rare states whose required controls determine the full-set cost. At the same time, an approximately feasible input need not follow a selected exact code. It may switch between overlapping feasible branches, leave the constraint within the permitted tolerance, and later return. A symbolic count alone does not settle the minimum physical catalogue.

The main question is whether the angular branch geometry and radial contraction must match for a common low-precision cost to exist. Theorem 2.1 answers this within a specified first-order family, under controlled feasibility of the actual triangle, nonsingularity almost everywhere in it, and a common norm inequality toward the origin. The radial factors are $r_i = a_i^{\beta_i}$. The matching condition $\beta_0 = \beta_1$ is necessary and sufficient. No radial autonomy, boundary-ray preservation, globally small derivatives, compact support, or full angular image for every feasible word is required.

2020 *Mathematics Subject Classification.* Primary 93C55, 37B40; Secondary 93D15, 37C05.

Key words and phrases. Outer invariance complexity, first-order matching, precision spectrum, noninjective control maps, initial-set dependence.

When the powers differ, Theorem 2.2 separates the full-constraint cost γD from $\gamma h/\chi_R$ on a single nearly full-area compact initial set, for every sufficiently small positive exponent. The strict difference follows from a relative entropy identity after the physical catalogue bounds have been proved. When the powers agree, Theorem 2.3 gives the complete full-set spectrum. At and below βh , every positive-area compact initial set has rate γ/β . Above it, one fixed nearly full-area compact set has rate h , whereas the full-set rate increases to $\log 2$. The distinct saturation boundary is $-(\beta/2)\log(a_0 a_1)$. In both cases the compact set is selected before all exponents, precision sequences, and horizons in the respective statement.

Two physical estimates give these results. Theorem 2.5 controls the area covered by any approximately feasible input. Near the origin, a logarithmic slope cone bounds complete feasible intervals without requiring them to map onto the full angular range. The angular product $\lambda_w = \prod a_{w_j}$ and radial product $P_w = \prod r_{w_j}$ enter separately: a first true exit gives a physical area band of size at most $C\delta\lambda_w/P_w$. The finite transient is handled on one regular compact subset of the initial set, where early true exits contribute only $O(\delta)$ area. Null critical sets permit null-set pullbacks along every feasible prefix. Finitely many local inverse charts then bound both early true exits and transported local areas, without an inverse Jacobian bound on the whole constraint. The compact subset is chosen before any precision sequence. The estimates apply to all feasible prefixes. For the full-set lower bound, Lemma 5.1 constructs actual bounded-run trajectories with a uniform margin against the alternative control. Forcing suitably shorter prefixes gives every frequency contribution to the spectrum before its physical margin falls below the tolerance. An auxiliary radial stopping tree constructs full-horizon upper catalogues, retaining only actual feasible assignments. In the matched case $P_w = \lambda_w^\beta$, and counting all leaf lengths and types evaluates the complete spectrum. Under mismatch, stopping lengths of typical words give the different near-full-area slope. The criterion extends these controlled estimates; it introduces no new pressure or graph-transform method.

Section 7 proves the mismatch bounds and the matching criterion for the same nonlinear first-order family. Section 8 gives an inward boundary example. In it, long constant words have empty feasible slices at each fixed positive radius, although every finite word is realised at sufficiently small radius. A constraint-preserving simultaneous conjugacy cannot reduce this example to boundary-ray-preserving models. A second example folds a genuinely feasible branch inside the constraint while its critical set still has area zero. Section 9 gives a smooth collapse example explaining why local matching and all-input convergence alone do not suffice.

1.1. Related work. Our finite-horizon count is the outer invariance complexity of Chen and Zhong [1, Section 2.2], with $n + 1$ constrained times instead of n . Their framework already allows noninjective continuous control maps. The fixed-tolerance equivalence results concern bounded catalogue size and finite outer equi-invariability. They do not quantify the catalogue constants along a tolerance sequence linked to the horizon. Stabilization entropies [3, 4] study related information costs under prescribed decay or practical stabilization conditions. Here the task is approximate viability of the actual constraint at every time, rather than a prescribed decay estimate for each state.

Measure-theoretic invariance entropy and variational principles are developed in [10, 9]. Chen, Huang and Zhong [2] study entropy dimensions using specified invariant partition cylinders and time-power weights. Their measures need not be invariant. Those cylinder estimates do not themselves bound the physical sets covered by arbitrary approximate inputs at δ_n . This framework also allows noninjective control maps. The first-exit estimate and the finite local-inverse-chart bound in Theorem 2.5 supply the additional geometric connection.

Graph transforms and volume distortion are established methods. Kawan and Da Silva [7, Appendix] give volume estimates near partially hyperbolic all-time lifts. In the present class, every bounded bi-infinite trajectory is the origin: iterating the norm inequality from time $-k$ bounds its time-zero norm by θ^k times a fixed bound. The blow-up angular coordinate is singular there and is not a physical unstable bundle. Their fixed-radius volume estimates do not provide the joint tolerance estimate needed here. We use the graph-transform method to prove that estimate directly, including truncated feasible images.

Product weights, stopping scales, types and typical sets are classical tools [6, 5]. Yang, Chen, Yang and Zhou [11] establish cost-threshold pressure and Bowen-equation results for a given invariant partition. Such address counts still require a comparison with all approximate physical inputs before they can determine our catalogue. The contribution here is that comparison through actual area and physical witnesses, rather than a new pressure or type formula. The comparisons with [7, 11] use the published graph-transform appendix and the author versions arXiv:1711.01181v2 and arXiv:2309.01628v3, respectively. The estimates used below are proved in full.

2. THE CATALOGUE AND MAIN RESULTS

Let

$$Q = \{(s, z) \in \mathbb{R}^2 : 0 \leq s \leq 1, |z| \leq s\}, \quad \text{area}(Q) = 1.$$

For an infinite binary input $u = (u_1, u_2, \dots)$ set $F_u^0 = \text{id}$ and $F_u^k = F_{u_k} \circ \dots \circ F_{u_1}$. All binary inputs are allowed. Distances are Euclidean, and all logarithms are natural. For a nonempty compact $K \subset Q$ define

$$(2.1) \quad A_u(n, \delta; K) = \{x \in K : \text{dist}(F_u^k x, Q) < \delta \text{ for every } 0 \leq k \leq n\}.$$

Let $r_F(n, \delta; K)$ be the smallest number of infinite inputs whose sets (2.1) cover K . The assignment may depend on the actual initial state. Exit and reentry are allowed subject to every displayed constraint. Only the first n letters affect this count. Controlled feasibility makes it finite.

A precision sequence means $\delta_n > 0$, $\delta_n \rightarrow 0$ and $-\log \delta_n/n \rightarrow \gamma$ for a finite $\gamma \geq 0$. It need not be monotone. Fix

$$(2.2) \quad a_0, a_1 \in (0, 1), \quad a_0 + a_1 = 1, \quad a_0 \neq a_1, \quad \beta_0, \beta_1 > 1,$$

and set $r_i = a_i^{\beta_i}$, $t_i = r_i/a_i < 1$, $a_* = \min_i a_i$, $a^* = \max_i a_i$ and $h = -\sum_i a_i \log a_i$. Write

$$(2.3) \quad A_0 = \begin{pmatrix} r_0 & 0 \\ t_0 a_1 & t_0 \end{pmatrix}, \quad A_1 = \begin{pmatrix} r_1 & 0 \\ -t_1 a_0 & t_1 \end{pmatrix}.$$

The standing assumptions on the actual maps are:

- (1) $F_0, F_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ are C^2 maps, $F_i(0) = 0$ and $DF_i(0) = A_i$.
- (2) $Q \subset F_0^{-1}Q \cup F_1^{-1}Q$.
- (3) For some $\Lambda \geq 1$ and $0 < \theta < 1$, the norm $V(s, z) = \max\{\Lambda|s|, |z|\}$ satisfies

$$(2.4) \quad V(F_i x) \leq \theta V(x) \quad (x \in \mathbb{R}^2, i = 0, 1).$$

- (4) Each critical set

$$(2.5) \quad \mathcal{C}_i = \{x \in Q : \det DF_i(x) = 0\}$$

has area zero.

The second assumption is one-step controlled invariance; unrestricted concatenation yields an exactly feasible infinite input from each point of Q . The third concerns contraction toward the origin, not distances between two arbitrary trajectories. In particular it controls appended tails even outside Q . The fourth supplies noncollapse along feasible finite transients; it allows folds and requires no condition on critical points outside Q . Radial

motion may depend on angle. No accumulated distortion, word-area bound or catalogue comparison is assumed.

Both matched and mismatched subfamilies are nonempty. For the linear maps $\mathbf{A}_i$, the angular domains $[-1, a_0 - a_1]$ and $[a_0 - a_1, 1]$ cover every fibre. Put $t = \max_i t_i < 1$ and choose $\Lambda = 1 + 4t/(1 - t)$. Then $V(\mathbf{A}_i x) \leq \max\{\max_i r_i, t(1 + 1/\Lambda)\}V(x)$ globally, with factor less than one, and $\det \mathbf{A}_i = r_i^2/a_i > 0$.

For words write

$$\lambda_w = \prod_{j=1}^{|w|} a_{w_j}, \quad P_w = \prod_{j=1}^{|w|} r_{w_j},$$

with empty products one. Define

$$(2.6) \quad \chi_R = - \sum_i a_i \log r_i > h, \quad \gamma_0 = - \log \max_i r_i > 0, \quad r_0^D + r_1^D = 1.$$

The root D is unique in $(0, 1)$, since the left side decreases from two to zero and $r_0 + r_1 < 1$. Also $\chi_R \geq \gamma_0$.

Theorem 2.1 (First-order matching criterion). *For every fixed system satisfying the standing assumptions, the following property holds if and only if $\beta_0 = \beta_1$: there are $\gamma_1 > 0$ and a function $c : (0, \gamma_1) \rightarrow [0, \infty)$ such that, for every fixed positive-area compact $K \subset Q$, every $\gamma \in (0, \gamma_1)$, and every precision sequence of exponent γ ,*

$$\lim_n n^{-1} \log r_F(n, \delta_n; K) = c(\gamma).$$

Theorem 2.2 (Low-precision mismatch). *Suppose $\beta_0 \neq \beta_1$. For each $\zeta \in (0, 1)$ there is one compact $K_\zeta \subset Q$, of area greater than $1 - \zeta$, such that simultaneously for every $0 < \gamma < \gamma_0$ and every precision sequence of that exponent,*

$$(2.7) \quad \lim_n \frac{\log r_F(n, \delta_n; Q)}{n} = \gamma D, \quad \lim_n \frac{\log r_F(n, \delta_n; K_\zeta)}{n} = \gamma h / \chi_R < \gamma D.$$

The set is selected before the exponent, sequence, and horizon. Every other fixed positive-area compact $K \subset Q$ has lower rate at least $\gamma h / \chi_R$ in this range.

Write H_b for the binary entropy. In the matched case $\beta_0 = \beta_1 = \beta$, the angular cost and spectrum are

$$(2.8) \quad H_b(p) = -p \log p - (1 - p) \log(1 - p), \quad \chi(p) = -p \log a_0 - (1 - p) \log a_1,$$

$$(2.9) \quad S(\gamma) = \max_{p \in [0, 1]} H_b(p) \min\{1, \gamma / [\beta \chi(p)]\},$$

where $0 \log 0 = 0$. In particular $H_b(a_0) = \chi(a_0) = h$.

Theorem 2.3 (Precision spectrum and initial-set dependence). *Suppose $\beta_0 = \beta_1 = \beta$ in the standing assumptions. For every fixed system, every finite $\gamma \geq 0$ and every precision sequence of exponent γ ,*

$$(2.10) \quad \lim_n \frac{\log r_F(n, \delta_n; Q)}{n} = S(\gamma).$$

Every fixed positive-area compact $K \subset Q$ satisfies

$$(2.11) \quad \min\{h, \gamma/\beta\} \leq \liminf_n \frac{\log r_F(n, \delta_n; K)}{n} \leq \limsup_n \frac{\log r_F(n, \delta_n; K)}{n} \leq S(\gamma).$$

Thus the rate exists and is γ/β for $0 \leq \gamma \leq \beta h$. For each $\zeta \in (0, 1)$ there is one fixed compact $K_\zeta \subset Q$ with $\text{area}(K_\zeta) > 1 - \zeta$ such that, simultaneously for every finite $\gamma \geq 0$ and every precision sequence of that exponent,

$$(2.12) \quad \lim_n \frac{\log r_F(n, \delta_n; K_\zeta)}{n} = \min\{h, \gamma/\beta\}.$$

The set is chosen before the exponent, sequence and horizon.

Corollary 2.4 (Closed spectrum and two thresholds). *Under the hypotheses of Theorem 2.3, put $\alpha = a_* \in (0, 1/2)$, $c_b = -\log(1 - \alpha)$ and $d_b = \log((1 - \alpha)/\alpha) > 0$. The common-cost boundary and saturation boundary are respectively*

$$(2.13) \quad \gamma_* = \beta h, \quad \gamma_{\text{sat}} = -\frac{\beta}{2} \log(a_0 a_1) = \beta(c_b + d_b/2) > \gamma_*.$$

The full-set spectrum is

$$(2.14) \quad S(\gamma) = \begin{cases} \gamma/\beta, & 0 \leq \gamma \leq \gamma_*, \\ H_b((\gamma/\beta - c_b)/d_b), & \gamma_* < \gamma < \gamma_{\text{sat}}, \\ \log 2, & \gamma \geq \gamma_{\text{sat}}. \end{cases}$$

The exact gap between Q and the single set K_ζ in the theorem is

$$(2.15) \quad \Delta(\gamma) = S(\gamma) - \min\{h, \gamma/\beta\} = \begin{cases} 0, & 0 \leq \gamma \leq \gamma_*, \\ H_b((\gamma/\beta - c_b)/d_b) - h, & \gamma_* < \gamma < \gamma_{\text{sat}}, \\ \log 2 - h, & \gamma \geq \gamma_{\text{sat}}. \end{cases}$$

It is strictly positive exactly when $\gamma > \gamma_*$. The full-set rate equals $\log 2$ exactly when $\gamma \geq \gamma_{\text{sat}}$.

The first threshold concerns equality of rates across all positive-area compact initial sets; the second concerns the full-set catalogue alone. The theorem does not classify high-side rates of arbitrary compact initial sets or assert that only two such rates are possible.

Theorem 2.5 (Area covered by an arbitrary approximate input). *Under the standing assumptions, whether matched or mismatched, there is a fixed integer $J \geq 1$ with the following property. For every positive-area compact $K \subset Q$ and every $0 < \varepsilon < \text{area}(K)$, one can choose a compact $T \subset K \setminus \{0\}$ with $\text{area}(T) > \text{area}(K) - \varepsilon$ and $C_T < \infty$ so that for every $e \in (0, h)$ there is $C_{T,e} < \infty$ such that*

$$(2.16) \quad \text{area}(T \cap A_u(n, \delta; K)) \leq C_T \delta + C_{T,e} \left[e^{-(h-e)m} + \delta e^{(\chi_R - h + 2e)m} \right]$$

for every input u , $n \geq J$, $0 \leq m \leq n - J$ and $0 < \delta \leq 1$. The integer depends only on the fixed system. The set and C_T precede e, n, δ, u ; the second constant may also depend on e . In particular, for $K = Q$ the set can have area greater than $1 - \zeta$ for any prescribed $\zeta \in (0, 1)$.

3. LOCAL FEASIBLE GEOMETRY

Use (s, y) with $z = sy$, only for $s > 0$. Define

$$R_i(s, y) = F_i^s(s, sy)/s, \quad N_i(s, y) = F_i^z(s, sy)/s, \quad T_i = N_i/R_i.$$

The integral formula $F_i(s, sy)/s = \int_0^1 D F_i(ts, tsy)(1, y) dt$ extends R_i, N_i as C^1 functions to $s = 0$. At zero

$$(3.1) \quad R_i(0, y) = r_i, \quad L_0(y) := T_0(0, y) = (y + a_1)/a_0, \quad L_1(y) := T_1(0, y) = (y - a_0)/a_1.$$

We choose the local constants from the maps as follows. Let L bound the Euclidean bilinear operator norm of both second derivatives on $\overline{B}(0, 2)$, and let $r_* = \min r_i$, $r^* = \max r_i$, $a^* = \max a_i$, $a_* = \min a_i$. Take $B = 10^4(1 + L)/r_*^3$, and shrink $\rho > 0$ so that

$$(3.2) \quad \rho \leq \min \left\{ \frac{1}{4}, \frac{r_*}{8(1 + L)}, \frac{1 - r^*}{8B}, \frac{1/a^* - 1}{16B}, \frac{1}{16B}, \frac{a_*}{16B} \right\}.$$

Taylor's integral formula gives $|R_i - r_i| \leq 2Ls$, $|\partial_s R_i|, |\partial_s N_i| \leq 2L$, and $|\partial_y R_i| \leq 2Ls$, $|\partial_y N_i - t_i| \leq 2Ls$. The same bounds apply with the appropriate constant linear term in N_i .

In particular $R_i \geq r_*/2$. Dividing these formulas by R_i , and differentiating the quotient, gives, with the generous common constant B ,

$$(3.3) \quad \begin{aligned} |R_i - r_i| &\leq Bs, & |\log(R_i/r_i)| &\leq Bs, \\ |\partial_s \log R_i| &\leq B, & |\partial_y \log R_i| &\leq Bs, \\ |T_i - L_i| &\leq Bs, & |\partial_s T_i| &\leq B, & |\partial_y T_i - 1/a_i| &\leq Bs. \end{aligned}$$

For example $|N_i| \leq 3$ on this rectangle; quotient derivatives use $R_i^{-2} \leq 4/r_*^2$. Thus the displayed choice of B dominates all the constants in (3.3). These are derived local inequalities, not extra hypotheses.

Set $\bar{r} = r^* + B\rho < 1$ and $\hat{r} = r^* + 2B\rho < 1$. On an exactly feasible local orbit, $s_k \leq \bar{r}^k s_0$. The bounds in (3.2) also give $T_1(s, -1) < -1$ and $T_0(s, 1) > 1$. Controlled invariance therefore forces

$$(3.4) \quad T_0(s, -1) \geq -1, \quad T_1(s, 1) \leq 1.$$

Equality is not required in (3.4). The increasing functions $T_i(s, \cdot)$ have unique cut points $\vartheta_0(s)$ with $T_0 = 1$ and $\vartheta_1(s)$ with $T_1 = -1$. They equal $a_0 - a_1 + O(s)$. Their feasible intervals are respectively $[-1, \vartheta_0(s)]$ and $[\vartheta_1(s), 1]$. Controlled invariance gives $\vartheta_1 \leq \vartheta_0$. Thus any local overlap has angular width $O(s)$ and physical width $O(s^2)$. Positivity of the overlap is not assumed.

3.1. Complete intervals with truncated images. Consider a radial graph $s = S(y)$, on any closed subinterval of $[-1, 1]$, with $|S'/S| \leq 1$. Put $k = S'/S$. For either input its terminal angular derivative along the graph is

$$J_i = \partial_y T_i + \partial_s T_i S' = 1/a_i + O(BS) > 0.$$

The new radial graph has logarithmic slope

$$(3.5) \quad k_{\text{new}} = \frac{k + (\partial_s \log R_i)S' + \partial_y \log R_i}{J_i}.$$

Its numerator is at most $1 + 2B\rho$ in absolute value and its denominator at least $1/a^* - 2B\rho$. By (3.2), $|k_{\text{new}}| \leq 1$. Piecewise C^1 graphs obey the same assertion almost everywhere.

For a fixed initial radius $0 < s_0 \leq \rho$, start with the horizontal graph $S \equiv s_0$. Let $I_w(s_0)$ be the set of initial angles whose trajectory under a finite word w stays in Q at every intermediate time. It is a closed interval, possibly empty or a point. Here is the induction that checks all constraints. The current angular image is an interval, with a graph obeying (3.5). Extend that graph constantly to the two sides of its domain. The extension is positive, bounded by ρ , and obeys the logarithmic-slope bound almost everywhere. Both $T_i(S(y), y)$ are increasing. By (3.4), control 0 cannot violate the lower angular boundary, and control 1 cannot violate the upper one. The other boundary imposes one cut. Pulling that cut back through the increasing preceding angular map leaves an interval. No terminal map is asserted to be onto $[-1, 1]$. The two allowed children cover the parent, by controlled invariance, but either child can be empty.

Along any such feasible graph, (3.3) and the geometric decay of s_j give constants independent of w, s_0 and its initial angle:

$$(3.6) \quad C_R^{-1} s_0 P_w \leq s_{|w|} \leq C_R s_0 P_w, \quad C_A^{-1} \lambda_w^{-1} \leq \frac{dY_w}{dy_0} \leq C_A \lambda_w^{-1}.$$

For instance one may take $C_R = \exp(B\rho/(1 - \bar{r}))$, $C_A = \exp(4B\rho/(1 - \bar{r}))$: $|a_i J_i - 1| \leq 2Bs_j \leq 1/8$, so its logarithm has modulus at most $4Bs_j$. The first estimate sums $\log(R_i/r_i)$; the second sums the angular derivatives. They are independent product estimates. Only in the matched case is $P_w = \lambda_w^\beta$. Integration over the *actual* angular image implies

$$(3.7) \quad |I_w(s_0)| \leq 2C_A \lambda_w, \quad |\mathcal{I}_w| \leq C_A \rho^2 \lambda_w, \quad \mathcal{I}_w = \{(s, sy) : 0 < s \leq \rho, y \in I_w(s)\}.$$

The Jacobian of $(s, y) \mapsto (s, sy)$ is s . There is no positive lower bound on $|I_w|$; some fixed-radius slices in Section 8.1 are empty.

For a point interval the length bound is zero. Derivative comparisons on nondegenerate intervals extend to their endpoints by continuity; degenerate fibres remain in all covers and do not contribute area.

3.2. The first true exit. Suppose the prefix w of length $j - 1$ is exactly feasible and the next prescribed input i is the first one to leave Q . On the current graph the map T_i is increasing with derivative at least $1/(2a_i)$. The radial output is at least cs_0P_w , by (3.3), (3.6), with $c > 0$ fixed. If this wrong step nonetheless has distance less than δ from Q , then

$$(3.8) \quad (|z_j| - s_j)_+ < 2\delta.$$

This follows from the Euclidean Lipschitz linear functionals $z - s$ and $-z - s$. The output radius is positive and at most ρ , so the violated constraint is angular, not the upper radius. Consequently the wrong part of the terminal angular interval has diameter at most $C\delta/(s_0P_w)$. By (3.6), its initial-angle preimage has length at most

$$C\delta\lambda_w/(s_0P_w).$$

After integration against $s_0 ds_0$, its physical area is at most

$$(3.9) \quad C\delta\lambda_w/P_w.$$

This argument also covers a point interval or a terminal interval lying wholly beyond the cut: monotonicity bounds the diameter of the part satisfying (3.8), without requiring a cut point inside the actual image. It does not require the wrong trajectory to remain outside afterwards. Any later reentry cannot erase constraint (3.8) at its first exit. A different exact feasible coding is not counted as an exit and has no role in (3.9).

4. FINITE TRANSIENTS AND ACTUAL COVERED AREA

For a finite word w , let

$$(4.1) \quad D_w = \{x \in Q : F_{w|_k}x \in Q \ (1 \leq k \leq |w|)\}, \quad D_\emptyset = Q.$$

These sets are compact. Write w^- for the word with its last letter removed. At a first true exit under w , all derivative inputs still lie in Q , so the relevant domain is D_{w^-} . The last output need not belong to Q .

Lemma 4.1 (Null pullbacks along feasible prefixes). *For every Lebesgue null set $B \subset \mathbb{R}^2$, every control i , and every finite word w ,*

$$(4.2) \quad \text{area}(Q \cap F_i^{-1}B) = 0, \quad \text{area}(D_w \cap F_w^{-1}B) = 0.$$

For each nonempty word the compact set

$$(4.3) \quad Z_w = D_{w^-} \cap \{x : \det DF_w(x) = 0\}$$

also has area zero.

Proof. The inverse function theorem supplies an open inverse chart at every point of $Q \setminus \mathcal{C}_i$. A countable family of these charts covers that set. Exhaust each image by relatively compact subcharts; its C^1 inverse is Lipschitz on sufficiently small compact neighbourhoods. A Lipschitz map preserves null sets: cover a bounded null set by squares with arbitrarily small total area, and cover their images by discs whose total area is bounded by a constant times that sum. Thus every chart pullback of B is null. The omitted $\mathcal{C}_i$ is null by (2.5). A Borel null superset handles an arbitrary Lebesgue null set.

For $w = vi$, exact feasibility gives

$$D_w \cap F_w^{-1}B \subset D_v \cap F_v^{-1}(Q \cap F_i^{-1}B).$$

Induction proves the second assertion of (4.2), including the empty word. Finally, on D_{w-} the chain rule expresses $\det DF_w$ as a product of determinants evaluated at states in Q . If the factor at time j vanishes, the initial state belongs to

$$D_{w|j-1} \cap F_{w|j-1}^{-1} \mathcal{C}_{w_j}.$$

This is null by (4.2). Their finite union contains Z_w , which is compact by continuity and closedness of the feasibility domain. $\square$

Choose $J \geq 1$, depending only on the fixed system, so that

$$(4.4) \quad \theta^J \sup_Q V \leq \rho.$$

Every exactly feasible J -prefix ends in $Q_\rho = Q \cap \{s \leq \rho\}$.

Lemma 4.2 (One regular compact window). *For every positive-area compact $K \subset Q$ and $0 < \varepsilon < \text{area}(K)$, there is a compact $T \subset K \setminus \{0\}$, with $\text{area}(T) > \text{area}(K) - \varepsilon$, and $L_T < \infty$ such that*

$$(4.5) \quad \text{area}(T \cap D_{w-} \cap F_w^{-1} B) \leq L_T \text{area}(B)$$

for all $1 \leq |w| \leq J$ and all Lebesgue measurable $B \subset \mathbb{R}^2$. On the same set, for every $e \in (0, h)$ there is $D_e \geq 1$ such that every local exactly feasible tail v , after any exactly feasible J -prefix from T , satisfies

$$(4.6) \quad \begin{aligned} D_e^{-1} e^{-(h+e)|v|} &\leq \lambda_v \leq D_e e^{-(h-e)|v|}, \\ D_e^{-1} e^{-(\chi_R+e)|v|} &\leq P_v \leq D_e e^{-(\chi_R-e)|v|}. \end{aligned}$$

The set and L_T precede e , every input, tolerance and horizon.

Proof. For $d > 0$, the total weight $\sum \lambda_v$ of length- m words with $|\#0/m - a_0| > d$ is at most $2e^{-2d^2m}$. Indeed, for a Bernoulli variable X of mean a_0 , the second derivative of $\log \mathbb{E} e^{t(X-a_0)}$ is a tilted variance, at most $1/4$. Its value and derivative at zero vanish, so it is at most $t^2/8$. Exponential Markov with $t = 4d$ and $t = -4d$ gives the bound. By (3.7), the unions of the corresponding complete local feasible regions have summable area. Borel–Cantelli for rational $d > 0$ gives a Borel null set $B_{\text{loc}} \subset Q_\rho$, including the origin, outside which all feasible length- m prefixes have zero frequency tending uniformly to a_0 . Overlaps do not invalidate this union bound. Controlled invariance supplies at least one feasible prefix at every length.

Lemma 4.1 shows that

$$B_J = \bigcup_{|\sigma|=J} (D_\sigma \cap F_\sigma^{-1} B_{\text{loc}})$$

is null. The finite union $Z_J = \bigcup_{1 \leq |w| \leq J} Z_w$ is a compact null set. On $Q \setminus B_J$, the maximum frequency deviation over all feasible transient choices and all their feasible length- m tails tends to zero. It is measurable: for each m there are finitely many words and their complete feasibility domains are closed. Egorov on K , followed by removal of $B_J \cup Z_J \cup \{0\}$ and compact inner approximation, gives T of the stated area with uniform convergence. Allocate less than $\varepsilon/2$ loss to Egorov and less than $\varepsilon/2$ to inner approximation. Both logarithmic products are affine functions of frequency. Uniform convergence gives (4.6); finitely many early lengths are absorbed in D_e . This selects one set before all precision choices, rather than a set for each exponent.

For a word w with $1 \leq |w| \leq J$, the compact set $M_w = T \cap D_{w-}$ is disjoint from Z_w . The composition F_w is therefore regular at every point of M_w . Cover it by finitely many open inverse charts $U_{w,k}$, with compact closures inside regular inverse charts. Change of

variables on each chart gives

$$\text{area}(M_w \cap U_{w,k} \cap F_w^{-1}B) \leq \left(\sup_{\bar{U}_{w,k}} |\det DF_w|^{-1} \right) \text{area}(B).$$

The supremum is finite. Sum over the charts and take the maximum over the finite word family, omitting empty M_w 's, to obtain (4.5). Chart overlaps only enlarge the bound; every local inverse branch meeting M_w is included. $\square$

The lemma provides an inverse-area bound on a fixed regular window, not a Jacobian lower bound on all Q or a global multiplicity bound. Its constants may depend on the prescribed area loss. The upper catalogues below still cover every actual state, including the critical states omitted from this window.

Proof of Theorem 2.5. Choose J, T, L_T from Lemma 4.2. Let $Q_\delta = \{x : \text{dist}(x, Q) < \delta\}$. For $0 < \delta \leq 1$, the tubes around the three edges of Q give

$$\text{area}(Q_\delta \setminus Q) \leq C_Q \delta, \quad C_Q = 2(2 + 2\sqrt{2}) + 3\pi.$$

If a state in $T \cap A_u(n, \delta; K)$ first truly exits at $j \leq J$, put $w = u|j$. Its earlier states lie exactly in Q , and it belongs to

$$T \cap D_{w-} \cap F_w^{-1}(Q_\delta \setminus Q).$$

By (4.5), its class has area at most $L_T C_Q \delta$. Summing over the J possible first-exit times gives

$$(4.7) \quad \text{area}\{x \in T \cap A_u(n, \delta; K) : \text{first true exit occurs by } J\} \leq C_T \delta, \quad C_T = J L_T C_Q.$$

All derivative inputs here are in Q ; no nonsingularity condition is needed at the last output, which may be outside.

For the remaining states, $\sigma = u|J$ is exactly feasible. Transport them by F_σ to Q_ρ , and use (4.5) to pull back endpoint area estimates. If the next m prescribed digits are exactly feasible for at least one transported state, (3.7) and (4.6) bound the area of the survival class by $C_{T,e} e^{-(h-e)m}$. If no such state exists, that class is empty. For a first true local exit at time $j \leq m$, the prescribed $(j-1)$ -prefix is feasible for a transported state whenever the class is nonempty. Its pulled-back area, by (3.9), (4.5) and both products in (4.6), is at most

$$C_{T,e} \delta e^{(\chi_R - h + 2e)(j-1)}.$$

Since $\chi_R > h$, the sum is bounded by a fixed multiple of $\delta e^{(\chi_R - h + 2e)m}$. Together with (4.7) and exact survival, this proves (2.16). A switch between feasible codes was never counted as an exit. Every covered state satisfies the constraint at its first true exit, regardless of subsequent reentry. $\square$

5. PHYSICAL TRAJECTORIES WITH A WRONG-CONTROL MARGIN

Lemma 5.1 (Bounded-run realisation). *For each integer $L \geq 2$, there is a positive radius s_L and $d'_L > 0$ such that every infinite binary sequence whose successive runs have length at most L is realised by an actual exactly feasible orbit starting at $(s_L, s_L y_0)$. At every step of that orbit the alternative control is genuinely infeasible, with an angular excess at least d'_L . The radius and margin are uniform over the sequences. No such assertion is made for unbounded runs or all constant words.*

Proof. The inverse reference maps are $f_0(y) = a_0 y - a_1$, $f_1(y) = a_1 y + a_0$. For any fixed infinite input v , successive inverse compositions give a unique reference bounded orbit $y_k^0 \in [-1, 1]$. If runs have length at most L , the next opposite digit is seen within $L + 1$

digits. Since $f_0(y) + 1 = a_0(y + 1)$, $1 - f_1(y) = a_1(1 - y)$, and each opposite digit inserts an endpoint distance at least $2a_*$,

$$(5.1) \quad |y_k^0| \leq 1 - d_L, \quad d_L = 2a_*^{L+1} > 0.$$

Let $b = d_L/4$ and consider the complete closed ball of sequences $\|y - y^0\|_\infty \leq b$ in ℓ^∞ . For each such sequence and a positive fixed initial radius s_* , recursively define $s_{k+1} = s_k R_{v_{k+1}}(s_k, y_k)$. These radii stay below $\bar{r}^k s_*$, independently of whether the angular recursion has yet been imposed. The scalar map $s \mapsto s R_i(s, y)$ has derivative bounded in absolute value by $\hat{r} < 1$, and its y derivative is bounded by Bs^2 , since $R_i \leq \bar{r} < 1$ and (3.3) bounds its logarithmic angular derivative. Thus two candidate sequences give

$$(5.2) \quad \sup_k |s_k(y) - s_k(\tilde{y})| \leq C_s s_*^2 \|y - \tilde{y}\|_\infty, \quad C_s = B/(1 - \hat{r}).$$

Write $e_i(s, y) = T_i(s, y) - L_i(y)$. By (3.3), $|e_i| \leq Bs$, $|\partial_s e_i| \leq B$, and $|\partial_y e_i| \leq Bs$. Define

$$(5.3) \quad (\mathcal{B}y)_k = f_{v_{k+1}}(y_{k+1}) - a_{v_{k+1}} e_{v_{k+1}}(s_k(y), y_k).$$

Choose $s_* \leq \rho$ so small that

$$(5.4) \quad a^* B s_* \leq (1 - a^*)b/2, \quad a^* B(s_* + C_s s_*^2) < (1 - a^*)/2.$$

Then (5.3) maps the ball into itself and has Lipschitz constant at most $a^* + a^* B(s_* + C_s s_*^2) < 1$. The contraction mapping theorem gives a fixed sequence. Its equation is exactly $y_{k+1} = T_{v_{k+1}}(s_k, y_k)$, not a formal symbolic substitution. It has $|y_k| \leq 1 - 3d_L/4$, so the constructed physical trajectory is exactly feasible for all times.

To check the alternative control, suppose the chosen digit is zero. Then $T_0(s_k, y_k) = y_{k+1} \leq 1 - 3d_L/4$, and (3.3) gives

$$y_k \leq a_0 - a_1 - 3a_0 d_L/4 + a_0 B s_k.$$

Substitution into $T_1 = L_1 + e_1$ yields

$$T_1(s_k, y_k) \leq -1 - \frac{3a_0 d_L}{4a_1} + B(1 + a_0/a_1)s_k.$$

For chosen digit one, the symmetric computation gives

$$T_0(s_k, y_k) \geq 1 + \frac{3a_1 d_L}{4a_0} - B(1 + a_1/a_0)s_k.$$

Shrink s_* once more, before considering any particular sequence, so that these last error terms are at most $a_* d_L/(4a^*)$. Both wrong angular excesses are then at least $d'_L = a_* d_L/(2a^*)$. Set $s_L = s_*$. All choices depend only on L and the fixed system. This proves the lemma, including the physical realisation and the genuine-infeasibility assertion. $\square$

Lemma 5.2 (Forcing a block prefix). *Fix $L \geq 2$ and $1 \leq j \leq L - 1$, and put*

$$M_L = \binom{L-2}{j-1}, \quad c_L = -\frac{j}{L} \log r_0 - \left(1 - \frac{j}{L}\right) \log r_1.$$

For every precision sequence of exponent $\gamma > 0$ and every $0 < t < \min\{1, \gamma/c_L\}$, for all sufficiently large n ,

$$(5.5) \quad r_F(n, \delta_n; Q) \geq M_L^{\lfloor tn/L \rfloor}.$$

Proof. Use all length- L blocks starting with zero, ending with one, and containing j zeros. Every concatenation has runs of length at most L and radial product e^{-Lc_L} per complete block. Put $k_n = \lfloor tn/L \rfloor$ and $m_n = Lk_n$. For each of the $M_L^{k_n}$ initial concatenations, append one fixed allowable block forever. Lemma 5.1 realises these infinite inputs at the same positive radius s_L . Every prefix through time m_n has $P_{v|k} \geq P_{v|m_n} = e^{-c_L m_n}$.

If an arbitrary catalogue input first differs from a witness at $j' \leq m_n$, its preceding states agree with that witness. By (3.6), their radius at $j' - 1$ is at least $C_R^{-1} s_L e^{-c_L m_n}$.

The wrong radial factor is at least $r_*/2$, and the alternative angular excess is at least d'_L . Its actual physical excess therefore obeys

$$(5.6) \quad |z_{j'}| - s_{j'} \geq E_L e^{-c_L m_n}, \quad E_L = \frac{s_L r_* d'_L}{2C_R} > 0.$$

The ratio to $2\delta_n$ has logarithmic rate $\gamma - c_L t > 0$. For large n , it violates (3.8), regardless of later reentry. Thus the approximate input must agree with all m_n witness digits. Distinct selected prefixes give distinct actual states: at their first different digit one input is feasible and the other has the proved wrong-control margin. One catalogue input covers at most one of them, proving (5.5). The block length, positive radius, and margin remain fixed throughout this horizon limit. Integer rounding and every $o(n)$ precision fluctuation are absorbed by the strict choice of t . $\square$

6. STOPPING CATALOGUES AND THE MATCHED SPECTRUM

We first give a catalogue construction under the standing assumptions. The reference tree provides an upper bound; it is not presumed necessary or optimal.

Lemma 6.1 (A complete radial stopping catalogue). *Put $M = \max\{1, 4\sqrt{2}\Lambda C_R \rho\}$. For $N \geq 0$ and $0 < t \leq 1$, let $\mathcal{L}_P(N, t)$ be the full reference binary tree stopped at the first $P_w \leq t$, or at depth N , whichever occurs first. For $n \geq J$ and $0 < \delta \leq 1$,*

$$(6.1) \quad r_F(n, \delta; Q) \leq 2^J \# \mathcal{L}_P(n - J, \delta/M).$$

The catalogue can retain only transient/leaf pairs whose complete actual feasible initial region is nonempty.

Proof. For every actual initial state choose an exactly feasible infinite input. Its first J letters form a feasible transient ending in Q_ρ ; the subsequent letters reach a reference leaf. Keep every pair realised by an actual state, including singleton regions and boundary states. Before stopping the assigned states are exactly feasible. At a threshold leaf, (3.6) gives

$$s_{|w|} \leq C_R \rho P_w \leq C_R \rho \delta/M \leq \delta/(4\sqrt{2}\Lambda).$$

Since $V = \Lambda s$ on Q , appending the same infinite zero tail and using (2.4) keeps the Euclidean norm at most $\delta/4$ at every later time. The origin belongs to Q , so every constraint holds strictly on that tail. A horizon leaf is exactly feasible through time n , and its later extension is irrelevant. Every constructed input fixes the origin as well. There are at most 2^J transients, proving (6.1). $\square$

6.1. Upper bound for the matched spectrum. For the rest of this section assume $\beta_0 = \beta_1 = \beta$. Then $P_w = \lambda_w^\beta$, $\chi_R = \beta h$, and

$$t_\delta = (\delta/M)^{1/\beta}, \quad b_\delta = -\log t_\delta.$$

Lemma 6.1 uses the tree stopped at $\lambda_w \leq t_\delta$ or at depth $N = n - J$. Put $c_i = -\log a_i$, $c_{\min} = \min_i c_i > 0$, $c_{\max} = \max_i c_i$, and $\Gamma_{n,\delta} = \beta(b_\delta + c_{\max})/n$. A threshold leaf has logarithmic angular cost in $[b_\delta, b_\delta + c_{\max}]$; an unfinished horizon leaf has cost below b_δ . Thus every nonempty leaf of length $\ell \leq n$, containing j zeros with $p = j/\ell$, satisfies

$$(6.2) \quad \ell \chi(p) < b_\delta + c_{\max}, \quad \frac{\ell}{n} \leq \min\{1, \Gamma_{n,\delta}/[\beta \chi(p)]\}.$$

The corresponding term of the binomial law of parameter j/ℓ gives $\binom{\ell}{j} \leq e^{\ell H_b(j/\ell)}$. Endpoint types have one word. Each leaf type therefore has at most $e^{nS(\Gamma_{n,\delta})}$ words. There are at most $(n+1)^2$ length/type pairs, including a possible empty word. By the actual catalogue construction,

$$(6.3) \quad r_F(n, \delta; Q) \leq 2^J (n+1)^2 e^{nS(\Gamma_{n,\delta})}.$$

No nonempty fixed-radius slice or full terminal image is assumed.

Every function under the maximum defining S is Lipschitz in γ , with common constant at most $\log 2/(\beta c_{\min})$. Their maximum is continuous. Here $\beta b_{\delta_n} = -\log \delta_n + \log M$, so $\Gamma_{n,\delta_n} \rightarrow \gamma$ for every precision sequence. The fixed transient and polynomial factors have zero rate, giving

$$(6.4) \quad \limsup_n n^{-1} \log r_F(n, \delta_n; Q) \leq S(\gamma).$$

This includes $\gamma = 0$ and nonmonotone sequences.

The reference angular leaf weights sum to one, and each is at least $a_* t_\delta$: a threshold leaf has parent weight above t_δ , whereas an unfinished leaf has weight above t_δ itself. This identity concerns a reference tree, not a physical partition. Enlarging a constant for $n < J$ yields the useful coarse bound

$$(6.5) \quad r_F(n, \delta; Q) \leq C_U \delta^{-1/\beta} \quad (0 < \delta \leq 1), \quad r_F(n, \delta; Q) \leq 2^n.$$

6.2. Shorter physical prefixes give the lower bound. Let $\gamma > 0$ and fix $p \in (0, 1)$. Choose $j_L \in \{1, \dots, L-1\}$ with $j_L/L \rightarrow p$, and put

$$M_L = \binom{L-2}{j_L-1}, \quad B_L = L^{-1} \log M_L, \quad \chi_L = \chi(j_L/L).$$

The exact identity and elementary type bounds

$$\binom{L-2}{j-1} = \frac{j(L-j)}{L(L-1)} \binom{L}{j}, \quad \frac{e^{LH_b(j/L)}}{L+1} \leq \binom{L}{j} \leq e^{LH_b(j/L)}$$

give $B_L \rightarrow H_b(p)$ and $\chi_L \rightarrow \chi(p)$. For the lower type bound the j -th term is a largest term of the binomial law with parameter j/L , so is at least $1/(L+1)$. The additional fraction has logarithm $o(L)$ for interior p .

The radial block cost in Lemma 5.2 is now $c_L = \beta \chi_L$. Fix L first and take

$$(6.6) \quad 0 < t < \min\{1, \gamma/(\beta \chi_L)\}.$$

The lemma, which checks the physical margin and every constrained time, gives

$$(6.7) \quad \liminf_n n^{-1} \log r_F(n, \delta_n; Q) \geq t B_L.$$

For fixed L take the supremum over the strict t , next let $L \rightarrow \infty$, and finally take the supremum over $p \in (0, 1)$. Endpoint frequencies have zero entropy, so continuity yields $\liminf n^{-1} \log r_F(n, \delta_n; Q) \geq S(\gamma)$. No witness radius or margin is required to remain uniformly positive as $L \rightarrow \infty$; they are fixed in each horizon limit. For $\gamma = 0$, the bound $r_F \geq 1$ and (6.4) give the zero limit. This proves (2.10).

6.3. Positive-area and fixed near-full initial sets. To complete Theorem 2.3, apply Theorem 2.5 to a fixed positive-area compact K and its positive-area set T . A catalogue covering K has at least $\text{area}(T)/\sup_u \text{area}(T \cap A_u(n, \delta_n; K))$ elements. For fixed $e \in (0, h)$, put

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\chi_R + e} \right\rfloor \right\}.$$

Eventually $m_n \geq 0$, and $\delta_n \leq e^{-(\chi_R + e)m_n}$. Each term on the right of (2.16) is then at most a constant times $e^{-(h-e)m_n}$. Consequently

$$\liminf_n n^{-1} \log r_F(n, \delta_n; K) \geq (h - e) \min\{1, \gamma/(\chi_R + e)\}.$$

Since $\chi_R = \beta h$ here, letting $e \downarrow 0$ gives the lower bound in (2.11); for $\gamma = 0$ the bound $r_F \geq 1$ also suffices. The upper bound follows from $r_F(n, \delta; K) \leq r_F(n, \delta; Q)$ and (2.10). The evaluation of S below will give the claimed common low-side rate.

For $\zeta \in (0, 1)$ take the single near-full set T constructed in Section 4 and put $K_\zeta = T \cup \{0\}$. All exactly feasible length- n words meeting T have, after their J -digit transient, frequency uniformly tending to a_0 . If $d_N \rightarrow 0$ bounds this deviation for tails of length $N = n - J$, there are at most

$$2^J(N+1) \exp\left(N \max_{|p-a_0| \leq d_N} H_b(p)\right) = e^{(h+o(1))n}$$

such words. Every state of T has an exactly feasible prefix, so these words, extended by any infinite tail, cover T through time n . One additional input covers the origin. This gives an upper rate h for every precision sequence, without selecting a strategy. Bound (6.5) also gives upper rate γ/β . The lower bound already proved gives (2.12). The construction of T used all-prefix frequency convergence, removal of null critical pullbacks, and compact inner approximation; it preceded every exponent, precision sequence and horizon. This completes Theorem 2.3 subject to the following elementary evaluation.

6.4. Evaluation of the maximum.

Proof of Corollary 2.4. Reparameterise the maximum by the frequency q of the smaller branch $\alpha = a_*$. This only changes the optimisation variable, not the dynamics or control assignments. Then $\chi = c_b + d_b q$. Write $x = \gamma/\beta$. The entropy inequality

$$c_b + d_b q - H_b(q) = q \log(q/\alpha) + (1-q) \log((1-q)/(1-\alpha)) \geq 0$$

follows from $-\log v \geq 1 - v$ and extends to the endpoints by continuity; equality holds at $q = \alpha$. For $x \leq h$, the objective $H_b(q) \min\{1, x/(c_b + d_b q)\}$ is at most x and equals x at α .

For $h < x < c_b + d_b/2$ put $q_x = (x - c_b)/d_b \in (\alpha, 1/2)$. When $q \leq q_x$ the objective is $H_b(q) \leq H_b(q_x)$. When $q \geq q_x$ it is $x H_b(q)/(c_b + d_b q)$. This ratio decreases for $q > \alpha$: the numerator of its derivative is

$$D(q) = (c_b + d_b q) H'_b(q) - d_b H_b(q), \quad D(\alpha) = 0, \quad D'(q) = (c_b + d_b q) H''_b(q) < 0.$$

The maximum is therefore $H_b(q_x)$. Finally, for $x \geq c_b + d_b/2$, the value at $q = 1/2$ is $\log 2$, the universal entropy maximum. Since $\alpha < 1/2$, $c_b + d_b/2 > c_b + d_b \alpha = h$. The three pieces match at the thresholds, proving (2.13)–(2.14) and the low-side assertion of Theorem 2.3. Subtracting (2.12) gives (2.15). Strict increase of H_b on $(\alpha, 1/2)$ proves the stated strict gap and the exact saturation threshold. $\square$

Corollary 6.2 (Matched linear systems). *If $\beta_0 = \beta_1 = \beta$, the pair (A_0, A_1) has all the conclusions of Theorems 2.3 and 2.5.*

Proof. The angular maps are $(y+a_1)/a_0$ and $(y-a_0)/a_1$. Their feasible domains $[-1, a_0-a_1]$ and $[a_0-a_1, 1]$ cover the fibre. Put $t = \max_i t_i < 1$, choose $\Lambda = 1 + 4t/(1-t)$ and set $v = t(1 + 1/\Lambda) < 1$. Then $V(A_i x) \leq \max\{\max_i r_i, v\} V(x)$ globally. The remaining hypotheses are immediate. $\square$

7. MISMATCH AND THE COMMON-COST CRITERION

We now return to the standing assumptions with $\beta_0 \neq \beta_1$. The same physical estimates apply, but the angular and radial products no longer have a common-power identity.

Proof of Theorem 2.2. First consider the upper rate on Q . For every $0 < \gamma < \gamma_0$, all branches of the tree in Lemma 6.1 cross $P_w \leq \delta_n/M$ before depth $n - J$ for large n , since

$$P_w \leq e^{-\gamma_0(n-J)} < \delta_n/M \quad \text{at that depth.}$$

The ratio has logarithmic rate $\gamma - \gamma_0 < 0$, so the statement includes every $o(n)$ tolerance fluctuation. For the uncapped radial tree, write $N(t)$ for its leaf count. Replacing a parent by its children preserves $\sum P_w^D = 1$, and each leaf satisfies $r_* t < P_w \leq t$. Hence

$$(7.1) \quad t^{-D} \leq N(t) \leq (r_* t)^{-D}, \quad r_F(n, \delta_n; Q) \leq 2^J N(\delta_n/M).$$

This proves the upper rate γD through an actual full-time catalogue, without asserting that the reference tree is necessary.

For the lower rate put $p_* = r_0^D$, so $1 - p_* = r_1^D$, and $c_R(p) = -p \log r_0 - (1 - p) \log r_1$. Then

$$(7.2) \quad H_b(p_*) = Dc_R(p_*).$$

Choose $j_L/L \rightarrow p_*$ and use the blocks in Lemma 5.2, with $c_L = c_R(j_L/L)$ and $B_L = L^{-1} \log \binom{L-2}{j_L-1}$. The binomial calculation in Section 6 gives $B_L \rightarrow H_b(p_*)$ and $c_L \rightarrow c_R(p_*)$. For each fixed L , $c_L \geq \gamma_0 > \gamma$. Every strict $0 < t < \gamma/c_L < 1$ therefore gives lower rate tB_L . Take its supremum first, then let $L \rightarrow \infty$. Equation (7.2) gives the lower rate γD . The positive-radius witnesses and wrong-control margins are those of the common physical lemma, not midpoints of assumed full images. This proves the first limit in (2.7).

For any fixed positive-area compact K , take the set T in Theorem 2.5, and set

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\chi_R + e} \right\rfloor \right\}.$$

For large n it is nonnegative, and $\delta_n \leq e^{-(\chi_R + e)m_n}$. Every term of (2.16) is then bounded by a constant times $e^{-(h-e)m_n}$. Covering the positive area of T implies lower rate $\gamma(h - e)/(\chi_R + e)$, since $\gamma < \gamma_0 \leq \chi_R$. Let $e \downarrow 0$. This proves the asserted universal lower bound $\gamma h/\chi_R$.

For $K = Q$, choose the single set T of area greater than $1 - \zeta$ in the area proof, and put $K_\zeta = T \cup \{0\}$. Keep only stopped pairs in Lemma 6.1 realised on T . They cover it with the complete tails already constructed. Let $q_\delta = -\log(\delta/M)$ and $c_{\max} = -\log r_*$. Each retained tail of length ℓ has

$$q_\delta \leq -\log P_w < q_\delta + c_{\max}, \quad \ell \leq \frac{q_\delta + c_{\max} + \log D_e}{\chi_R - e},$$

by (4.6). At this length, $\lambda_w \geq D_e^{-1} e^{-(h+e)\ell}$, while all reference length- ℓ weights sum to one. There are at most $D_e e^{(h+e)\ell}$ such words. Sum over these lengths and the at most 2^J transients, and include one input for the origin. For fixed $e \in (0, h)$ this gives

$$(7.3) \quad \log r_F(n, \delta_n; K_\zeta) \leq \frac{h + e}{\chi_R - e} [q_{\delta_n} + c_{\max} + \log D_e] + O_e(\log(1 + q_{\delta_n})) + O_e(1).$$

Because $q_{\delta_n} = \gamma n + o(n)$, letting $e \downarrow 0$ gives upper rate $\gamma h/\chi_R$, matching the lower bound. The set was selected through uniform frequency convergence before e , the precision exponent, the sequence, and the horizon.

Finally let $b_i = r_i^D$. The classical relative-entropy identity is

$$D\chi_R - h = \sum_i a_i \log(a_i/b_i) \geq 0.$$

Concavity of the logarithm proves the inequality, with equality only when $b_i = a_i$ for both labels. Equality would require $\beta_0 D = \beta_1 D = 1$, contrary to mismatch. Thus $D > h/\chi_R$, completing the theorem. $\square$

Proof of Theorem 2.1. If $\beta_0 = \beta_1 = \beta$, Theorem 2.3 gives the common rate $c(\gamma) = \gamma/\beta$ on $(0, \beta h)$. If $\beta_0 \neq \beta_1$, Theorem 2.2 separates Q and one fixed nearly full-area compact set for every $0 < \gamma < \gamma_0$. Any interval adjacent to zero contains such an exponent, contradicting the common-rate property. $\square$

The criterion concerns precisely the first derivatives (2.3), nonsingularity almost everywhere in Q , controlled feasibility, and the common norm inequality. The proof does not extend this necessity statement to arbitrary feedback systems, classify high-precision mismatch rates, or compare a fixed strategy with the optimal catalogue.

8. APPLICATIONS OF THE CRITERION

The examples distinguish truncated local feasible images from folding during a finite transient. The final example in this section also exhibits angle-dependent radial motion under mismatch.

8.1. Inward boundary images and empty fixed-radius slices. Let $t = \max a_i^{\beta-1} < 1$, $\Lambda = 1 + 4t/(1-t)$, $v = t(1 + 1/\Lambda) < 1$, and fix

$$0 < c < \min\{1 - a^*, (1 - v)/v, 1\}, \quad g(s) = 1 - \frac{cs}{1 + s^2}.$$

Consider the single explicit family

$$(8.1) \quad \begin{aligned} F_0(s, z) &= (r_0 s, g(s)t_0(z + a_1 s)), \\ F_1(s, z) &= (r_1 s, g(s)t_1(z - a_0 s)). \end{aligned}$$

Both are global smooth diffeomorphisms: $g \in [1 - c/2, 1 + c/2]$ is positive, the first coordinate has the explicit inverse $s = s'/r_i$, and the second is an affine invertible function of z at that radius. Their first jets are (2.3). For the stated V ,

$$(8.2) \quad V(F_i x) \leq \max\{r^*, (1 + c/2)v\} V(x), \quad \max\{r^*, (1 + c/2)v\} < 1.$$

Thus the norm inequality is checked globally, not inferred from feasible trajectories.

For $0 < s \leq 1$, $a^* < g(s) < 1$. The complete one-step angular domains are

$$(8.3) \quad D_0(s) = [-1, a_0/g(s) - a_1], \quad D_1(s) = [a_0 - a_1/g(s), 1].$$

Their endpoints lie strictly inside $(-1, 1)$, and their overlap has angular width $1/g(s) - 1 > 0$, physical width $s(1/g(s) - 1) = O(cs^2)$ near zero. This proves controlled invariance and the actual overlap geometry. Theorem 2.3 applies.

Yet for every fixed $0 < s_0 \leq 1$, a sufficiently long all-zero word is infeasible for every initial angle. To verify this, write $s_k = r_0^k s_0$ and $q_k = y_k + 1$. Under zero input,

$$(8.4) \quad q_{k+1} = \frac{g(s_k)}{a_0} q_k + 1 - g(s_k).$$

The multiplier is uniformly greater than one, since $g(s) > a^*$, and $q_1 \geq 1 - g(s_0) > 0$ for $y_0 \geq -1$. The sequence eventually exceeds two, violating $y \leq 1$. Monotonicity means that this also holds for the smallest initial angle, hence for all of them. Thus a description in which every word has a nonempty full-image slice does not apply to this example.

This statement concerns $I_0^m(s_0)$ at a fixed radius, not the whole physical region $\mathcal{I}_0^m$. Choosing the radius smaller can make a given finite word feasible. Indeed any finite word admits a bounded-run infinite extension, and Lemma 5.1 realises that extension at a sufficiently small positive radius. Missing fixed-radius slices must not be confused with removing a word everywhere in Q .

There is also a constraint-preserving conjugacy obstruction. For (8.1), F_0 sends the positive lower edge of Q strictly inside Q , and its upper edge outside; F_1 does the symmetric thing. The image of the top edge has constant radius $r_i < 1$ and crosses the opposite angular boundary exactly once. Thus

$$(8.5) \quad F_i(\partial Q) \cap \partial Q \text{ consists of exactly two points.}$$

In contrast, for any target pair whose zero control sends the whole lower edge of Q into ∂Q and whose one control sends the whole upper edge into ∂Q , both corresponding intersections contain a continuum. A homeomorphism H of the plane with $H(Q) = Q$, simultaneously conjugating the two labelled controls (even allowing a label permutation), would map each intersection in (8.5) onto the corresponding target intersection. Finite versus continuum is impossible. Thus this example is not reducible to such a boundary-edge-preserving pair by a fixed Q -preserving simultaneous control homeomorphism, and in particular not by such

a C^1 or bi-Lipschitz map. No assertion about unconstrained coordinate changes or every member of the new class is made.

For example, take $a_0 = 1/3$, $a_1 = 2/3$, $\beta = 2$ and $c = 1/100$. The choice $\Lambda = 9$ in (8.2) gives a global factor at most $67/90 < 1$. Corollary 2.4 evaluates this same inward family at exponent $\gamma = \log 4$:

$$2 \left(\log 3 - \frac{2}{3} \log 2 \right) < \log 4 < \log(9/2), \quad \lim_n n^{-1} \log r_F(n, \delta_n; Q) = H_b \left(\frac{\log(4/3)}{\log 2} \right).$$

This holds for every precision sequence of that exponent. The first inequality follows from $(4/3)^3 > 2$ and the second from $4 < 9/2$. It is an evaluation of the spectrum, not a claim that missing fixed-radius words require a different count.

8.2. A feasible fold with a null critical set. The next example shows that nonsingularity almost everywhere in Q genuinely allows noninjective feasible branches.

Proposition 8.1. *For every parameter choice (2.2), matched or mismatched, there are global smooth maps satisfying all standing assumptions which identify two distinct interior points of Q . For at least one control the common image belongs to Q . The maps have critical points in Q , but their critical sets there have area zero.*

Proof. Fix $c = 1/5$, let $E(v) = e^{-1/v}$ for $v > 0$ and zero otherwise, and put

$$\psi(s) = \frac{E(s - 1/4)}{E(s - 1/4) + E(1/2 - s)}.$$

The denominator is positive, $0 \leq \psi \leq 1$, $\psi = 0$ for $s \leq 1/4$, and $\psi = 1$ for $s \geq 1/2$. Define

$$(8.6) \quad H(s, z) = \begin{cases} (s, z), & s \leq 1/4, \\ (s, z + cs\psi(s) \sin(2\pi z/s)), & s > 1/4, \end{cases} \quad F_i = A_i H.$$

The quotient is used only away from $s = 0$; flatness at $s = 1/4$ gives a global smooth map. Near the origin H is the identity, so the required first derivatives agree.

On Q , the angular action of H is $g_\tau(y) = y + c\tau \sin(2\pi y)$, $\tau = \psi(s)$. For $0 \leq y \leq 1/2$, it lies in $[0, 1/2 + c]$; for $1/2 \leq y \leq 1$, it lies in $[1/2 - c, 1]$. Oddness handles negative angles, and $g_\tau(\pm 1) = \pm 1$. Thus $H(Q) \subset Q$. At every $H(x)$, one of the linear controls is feasible because their angular domains cover $[-1, 1]$. Their radial factors are less than one. Consequently Q is controlled invariant for F_i , including its boundary.

Put $m_* = \max\{r_0, r_1, t_0, t_1\} < 1$ and choose

$$\Lambda = 1 + \frac{2m_*(1 + 2c)}{1 - m_*}.$$

For every state of the plane, $|H^z - z| \leq c|s|$, hence

$$V(Hx) \leq (1 + c/\Lambda)V(x), \quad V(A_i x) \leq m_*(1 + 1/\Lambda)V(x).$$

It follows that (2.4) holds globally with

$$\theta = m_*(1 + 1/\Lambda)(1 + c/\Lambda) \leq m_*[1 + (1 + 2c)/\Lambda] < (1 + m_*)/2 < 1.$$

This controls arbitrary tails as well as exactly feasible orbits.

For $s > 1/4$,

$$(8.7) \quad \det DH = 1 + 2\pi c\psi(s) \cos(2\pi z/s), \quad \det DF_i = (r_i^2/a_i) \det DH.$$

At each fixed s , the determinant is either identically one or has finitely many zeros on $|z| \leq s$. Fubini therefore proves (2.5). At $s = 3/4$ it is positive at $y = 0$ and negative at $y = 1/2$, since $2\pi/5 > 1$, so actual critical points exist.

At this same radius write $g(y) = y + c \sin(2\pi y)$. Then

$$g(1/4) = 9/20 < 1/2, \quad g(2/5) = 2/5 + (1/5) \sin(\pi/5) > 1/2, \quad g(1/2) = 1/2.$$

The middle inequality uses $\sin(\pi/5) > \sin(\pi/6) = 1/2$. There is $y_1 \in (1/4, 2/5)$ with $g(y_1) = 1/2$. The distinct interior points $(3/4, (3/4)y_1)$ and $(3/4, 3/8)$ have the same image under both controls. At least one control is feasible at their common intermediate state $(3/4, 3/8)$, proving the claim. $\square$

Injectivity is preserved by coordinate conjugacy. These collisions exclude a simultaneous Q -preserving control homeomorphism to injective controls. Theorem 2.3 or 2.2 applies according to the chosen powers. The critical points show why the inverse-area estimate belongs on a regular compact window rather than on all Q .

8.3. Angular feedback in an admissible mismatched pair. Choose unequal β_0, β_1 in (2.2), and choose a nonnegative smooth cutoff κ equal to one on $\overline{B}(0, 3/2)$ and supported inside $B(0, 7/4)$. For example, with E as in (8.6), take

$$\kappa(x) = \frac{E(169/64 - |x|^2)}{E(169/64 - |x|^2) + E(|x|^2 - 9/4)}.$$

Its denominator is positive and its support has radius $13/8$. Put $\phi(s, z) = z^2 \kappa(s, z)$, $M_\phi = \max\{1, \|x \cdot \nabla \phi\|_\infty\}$, and fix $0 < c < 1/(2M_\phi)$. Define

$$H(x) = e^{-c\phi(x)}x, \quad F_i(x) = A_i H(x).$$

On each ray $x = te$, $t \geq 0$, the image radius has derivative $e^{-c\phi(te)}[1 - cte \cdot \nabla \phi(te)] > 0$ and tends to infinity. Thus H is bijective. Its derivative has determinant $e^{-2c\phi(x)}[1 - cx \cdot \nabla \phi(x)] > 0$; the inverse function theorem gives a smooth global inverse, including at zero. The map is the identity outside a compact set and $DH(0) = I$. Consequently $DF_i(0) = A_i$. Since the scalar factor is at most one, the linear norm inequality checked in Section 2 proves (2.4) globally. The angles are the reference maps (3.1), their domains cover each fibre, and the radius decreases. This proves controlled feasibility with all boundary states retained. On Q , $\phi = z^2$, so the first coordinate $r_i s e^{-cz^2}$ depends on the initial angle at fixed positive radius. Theorem 2.2 applies. This example verifies actual radial feedback; no assertion that it excludes every simultaneous coordinate reduction is made.

9. WHY LOCAL MATCHING ALONE IS INSUFFICIENT

The following example keeps the common attracting origin and even agrees with the linear maps on a neighbourhood of it. It collapses positive-area regions onto an invariant ray, violating the null-critical-set hypothesis.

Proposition 9.1. *For every matched choice $\beta_0 = \beta_1 = \beta$ in (2.2) and every $0 < \sigma < 1/2$, there are C^∞ maps $S_0, S_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with these properties:*

- (1) *They equal A_0, A_1 on $\{s \leq \sigma\}$, so all derivatives at the origin agree.*
- (2) *All inputs drive bounded sets uniformly exponentially to the origin, and Q is controlled invariant.*
- (3) *The compact set $K_\sigma = Q \cap \{s \geq 2\sigma\}$ has area $1 - 4\sigma^2$ and $r_S(n, \delta; K_\sigma) = 1$ for every n and every $\delta > 0$.*
- (4) *For every finite $\gamma \geq 0$ and every sequence $\delta_n \rightarrow 0$ with $-\log \delta_n/n \rightarrow \gamma$, the full-set cost satisfies*

$$\min\{h, \gamma/\beta\} \leq \liminf \frac{\log r_S(n, \delta_n; Q)}{n} \leq \limsup \frac{\log r_S(n, \delta_n; Q)}{n} \leq \min\{\log 2, \gamma/\beta\}.$$

In particular, for $0 < \gamma \leq \beta h$ the full-set rate is γ/β , but the positive-area set K_σ has rate zero.

Proof. Let $e(t) = 0$ for $t \leq 0$ and $e(t) = \exp(-1/t)$ for $t > 0$, and define

$$\psi(s) = \frac{e(s - \sigma)}{e(s - \sigma) + e(2\sigma - s)}.$$

The denominator is positive everywhere. Thus ψ is smooth, zero for $s \leq \sigma$, and one for $s \geq 2\sigma$. Put $\rho_i = a_i^\beta$ and define

$$(9.1) \quad S_0(s, z) = (\rho_0 s, (1 - \psi(s))a_0^{\beta-1}(z + a_1 s) - \psi(s)\rho_0 s),$$

$$(9.2) \quad S_1(s, z) = (\rho_1 s, (1 - \psi(s))a_1^{\beta-1}(z - a_0 s) + \psi(s)\rho_1 s).$$

They have the claimed local identity. Each $S_i(s, z)$ is a convex combination of $A_i(s, z)$ and $R_i(s, z)$, where $R_0 = (\rho_0 s, -\rho_0 s)$ and $R_1 = (\rho_1 s, \rho_1 s)$. All four linear maps contract toward zero in one adapted norm $\max\{M|s|, |z|\}$ for sufficiently large M : the second-coordinate factor for A_i is at most $a_i^{\beta-1} + a_i^{\beta-1}a_{1-i}/M < 1$, and R_i has factor at most ρ_i . Convexity of the norm gives the same contraction bound for S_i on all of $\mathbb{R}^2$.

For controlled feasibility let $t = \psi(s)$, $y = z/s$, and $\xi = a_0 - a_1$. If $t < 1$ the full feasible intervals are

$$J_0(s) = \left[-1, \min \left\{ 1, \xi + \frac{2a_0 t}{1-t} \right\} \right],$$

$$J_1(s) = \left[\max \left\{ -1, \xi - \frac{2a_1 t}{1-t} \right\}, 1 \right].$$

They follow by imposing $-1 \leq z'/s' \leq 1$ in (9.1)–(9.2), and their union is $[-1, 1]$. For $t = 1$ both branches are feasible on the whole fibre. The radius always decreases, so Q is controlled invariant.

On K_σ , one application of S_0 sends every state to the lower ray. At every radius $S_0(s, -s) = (\rho_0 s, -\rho_0 s)$, so input 0^∞ keeps the whole set exactly in Q . Its area is $\int_{2\sigma}^1 2s \, ds = 1 - 4\sigma^2$.

To obtain the full-set lower bound, use $Q_c = Q \cap \{s \leq c\}$ with $c = \sigma/2$. For every input, the radius of a state in Q_c stays at most c . Thus $\psi = 0$ forever along that trajectory, even when its angle leaves $[-1, 1]$, and its entire trajectory is exactly that of A_i . Consequently $r_S(n, \delta; Q) \geq r_A(n, \delta; Q_c)$. Corollary 6.2 and (2.11) give the lower bound.

For the upper bound choose an integer t_0 such that $(\max_i \rho_i)^{t_0} \leq \sigma$. Every initial state admits an exact feasible prefix of length t_0 , ending in the region $s \leq \sigma$. All subsequent trajectories there coincide with the linear system, for all inputs, since the radius continues to decrease independently of the angle. Concatenate all 2^{t_0} possible prefixes with a catalogue for the linear full set over the remaining horizon. For $n \geq t_0$ this gives

$$r_S(n, \delta; Q) \leq 2^{t_0} r_A(n - t_0, \delta; Q) \leq 2^{t_0} C_U \delta^{-1/\beta}.$$

The 2^n exact-prefix bound also holds. These yield the upper rates for every stated precision sequence. Finally $\det DS_i = (1 - \psi(s))a_i^{2\beta-1}$ vanishes on the saturated region, identifying the noninjective mechanism. $\square$

For fixed σ this proposition gives one large zero-cost initial set in one fixed system. Making its area arbitrarily close to one changes σ and the system. This differs from Theorem 2.3, where all K_ζ are chosen inside a single fixed system. The collapse example is a counterexample to sufficiency of local matching and uniform convergence. Its critical set has positive area, so it lies outside the standing assumptions. In particular the null target ray has a positive-area feasible pullback, unlike the fold in Proposition 8.1.

10. CONSEQUENCES AND SCOPE

Theorem 2.1 characterises the common low-precision cost in the specified first-order family. Unequal radial powers already separate the full triangle and one nearly full-area compact set at arbitrarily small positive exponents. The necessity statement permits angular feedback in the radius and has the same global hypotheses as the matched result.

Theorem 2.5 is the underlying estimate for actual initial states under every approximately feasible input. Its two products record different features: λ_w controls feasible angular size

and P_w controls the physical radius at a first true exit. Typicality of all feasible prefixes handles overlap, and the constraint at that first exit handles later reentry. On one fixed regular compact window, finite local inverse charts control all feasible transients and early true exits. Their constants may depend on the chosen area loss, but precede every precision sequence. No inverse Jacobian bound is required on the full triangle. No positive length or full image is required of each feasible slice.

The full-set lower bounds use rare actual states at a fixed small positive radius, with a margin against the wrong control. Typical positive-area mass instead gives the near-full-set slope. Under matching, the product identity and classical type counting give the complete spectrum and its two distinct thresholds. The auxiliary witness radius depends on the fixed block length; the near-full initial set precedes all stated precision choices.

The results retain binary planar dynamics, a particular triangular constraint, the displayed first-jet family, controlled feasibility, nonsingularity almost everywhere in Q , and contraction toward a common origin. They allow truncated feasible images and vanishing overlap, but do not treat constant overlap in the limiting angular maps. The feasible fold shows that global invertibility is unnecessary. The collapse example explains why local matching and all-input convergence alone are insufficient; it does not show that null critical sets are necessary for every possible cost theorem. The inward and folded examples prove only their stated constraint-preserving conjugacy obstructions. The high-precision spectrum under mismatch and the high-side rates of arbitrary compact initial sets are not classified. No fixed-strategy optimality or new pressure principle is asserted.

Generative AI disclosure. Generative AI was used extensively in developing mathematical arguments, checking proof chains and related literature, drafting and restructuring the English text, and producing the LaTeX source. These activities do not constitute verification by a human author. Author identity, affiliations, authorship approval and the scope of human mathematical review remain to be supplied and confirmed before any submission.

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